

Zero Is Not Nothing

Boundary-conditioned vacuum readouts, the Casimir effect,
and the registry errors of zero-point engineering

Governing discipline

A description can close a calculation without closing reality. A formal zero, an operational null, and an assertion of absolute absence remain different objects unless a declared transition carries one into another.

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EmergenceByDesign

Companion to Nested Transfer and Scale-Free Spectra. Lab notes — independent work; no claim of ownership or discovery.

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Abstract

The same words — *zero*, *vacuum*, *fluctuation*, *polarization*, *energy*, and *extraction* — name different mathematical objects in different zero-point-energy debates. This paper supplies a typed registry calculus for locating the resulting inferential gaps. It proves that cross-registry substitution requires an explicit type-preserving map; that refining an operational zero does not by itself produce an observer-independent absence; that a Casimir energy difference does not entail an absolute reservoir or a positive-work cycle; and that an experiment cannot discriminate interpretations that induce the same readout distribution. Standard Casimir, Lifshitz, Bogoliubov, dynamical-Casimir, and equilibrium-cycle calculations are retained. Competing vacuum-fluctuation, source-interaction, material-response, stochastic-electrodynamics, extraction-engineering, cosmological, and disclosure-adjacent positions are compared at their actual propositions rather than arranged as a hierarchy of reality. The method neither certifies nor prohibits unconventional engineering. It identifies the additional object, map, cycle, or comparator a claim must supply.

Standing and stopping condition

Tier I consists of standard mathematics, cited physical results, and the formal registry theorems proved here. Tier II reads relations after those objects are fixed. Tier III states research direction. The paper stops when each examined inference is either derived, contradicted, underdetermined, or ill-typed under a declared registry. It does not stop by declaring a final ontology of the vacuum.

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1. Origin: the quarrel hidden inside a glyph

Public argument over zero-point energy (ZPE) often appears to divide cleanly into proponents and skeptics. It does not. Within each camp, the same term may refer to a Hamiltonian ground-state contribution, a vacuum correlation function, a stochastic classical background, a regularization term, a boundary-dependent energy difference, a gravitational source, or an engineering reservoir. Opposing participants can therefore disagree intensely while never applying one proposition to one mathematical object.

The present cultural moment amplifies the compression. Government programs have historically commissioned speculative reports on vacuum-energy extraction and spacetime engineering; NASA's former Breakthrough Propulsion Physics program explicitly examined vacuum-fluctuation energy among high-risk propulsion questions; declassified Defense Intelligence Agency reports discuss both extraction concepts and metric engineering. Contemporary UAP-transparency activity supplies additional public attention. None of those institutional facts verifies ZPE extraction, recovered technology, or a vacuum-propulsion mechanism. Their relevance here is rhetorical: adjacency is frequently used as though it were a transition map from public disclosure to technical confirmation.

The paper refuses two shortcuts:

Casimir observation $\nRightarrow$ extractable absolute reservoir,

and

successful standard calculation $\nRightarrow$ ontological exhaustion.

The first blocks promotion by enthusiasm. The second blocks promotion by authority. Neither blocks a properly specified experiment.

Registry discipline

An official document demonstrates that a question was commissioned, recorded, or discussed. It does not inherit the truth of every physical premise considered inside it. An unresolved UAP report demonstrates unresolved classification under the available data. It does not select ZPE, metric engineering, or any other preferred mechanism.

2. The registry calculus

Definition 2.1 (Typed registry). A registry is a tuple

$$\mathcal{R} = (\Sigma, \{D_s\}_{s \in S}, I, \mathcal{A}, \mathcal{O}),$$

where Σ is a many-sorted language; D_s is the domain assigned to sort s ; I interprets the symbols; $\mathcal{A}$ gives the axioms and admissible transformations; and $\mathcal{O}$ gives the permitted observables and readouts.

Relevant sorts include states, operators, detector records, calibration classes, energy differences, work cycles, material media, and ontological assertions. A shared word does not erase these types.

Theorem 2.2 (Typed-substitution obligation). *Let $t : s$ and $u : s'$ with $s \neq s'$. Replacing t by u in a*

derivation is admissible only if a declared coercion

$$C_{s \rightarrow s'} : D_s \longrightarrow D_{s'}$$

is supplied and every predicate transported into the conclusion is preserved on the relevant domain.

Proof. In a many-sorted calculus, same-sort substitution is a formation rule. Cross-sort replacement is not well formed until a map produces a term of the target sort. If the conclusion uses predicate $P_{s'}$, validity additionally requires

$$P_s(x) \implies P_{s'}(C_{s \rightarrow s'}x)$$

for every relevant x . Lexical identity supplies neither the map nor predicate preservation. $\square$

Definition 2.3 (Registry transition and residue). A transition is a possibly partial map

$$T_{ij} : D_{\mathcal{R}_i} \rightharpoonup D_{\mathcal{R}_j}.$$

For predicates P_i, P_j , its transport residue is

$$\text{Res}_{T_{ij}}(P_i, P_j) = \{x \in \text{Dom } T_{ij} : P_i(x) \wedge \neg P_j(T_{ij}x)\} \cup (D_{\mathcal{R}_i} \setminus \text{Dom } T_{ij}).$$

Zero residue licenses the stated transport, not the fundamentality of either registry. Nonzero residue blocks that transport, not every possible successor map.

Definition 2.4 (Four-valued audit). For premises Γ and conclusion C ,

$$\text{Audit}(\Gamma, C) \in \{\text{Derived}, \text{Contradicted}, \text{Underdetermined}, \text{IllTyped}\}.$$

The last value applies when no well-formed transition carries the premises into the sort used by C .

This is not a many-valued ontology. It is a proof-status ledger inside a declared formal task.

3. Why an operational zero remains idealized

Definition 3.1 (Zero registration). A zero registration is

$$Z_{\mathcal{R}} = (O, C, q_{\text{ref}}, \delta, \Delta t, B, \mathbf{M}),$$

where O is an observable, C a calibration map, q_{ref} a comparator, δ a resolution or decision tolerance, Δt an observation interval, B boundary conditions, and $\mathbf{M}$ the measurement model. Its null acceptance set is

$$\mathcal{Z}_{\mathcal{R}} = \{x : |C(Ox) - q_{\text{ref}}| \leq \delta\}.$$

A reported zero communicates membership in a selected acceptance set. It is not generally an equivalence class: a finite-tolerance proximity relation need not be transitive. “No counts” means zero registered events during Δt under a declared detector, gate, and coupling; it does not communicate absence under every possible relation.

Proposition 3.2 (Refinement does not supply absolute absence). *Let $\delta_n \downarrow 0$ and let $\mathcal{Z}_{\mathcal{R}_n}$ be the corresponding null acceptance sets. Without compatibility, identifiability, and attainability assumptions, refinement*

does not imply that $\bigcap_n \mathcal{Z}_{\mathcal{R}_n}$ is a singleton, nonempty, physically attainable, registry-independent, or identical with ontological absence.

Proof. The acceptance sets need not be nested when calibrations, boundaries, or models vary. Even nested closed sets can have empty intersection outside compact settings. A singleton intersection is a set-theoretic result, not proof that a finite observer realizes the limiting operation. Finally, “ontological absence” is not an element of the operational codomain supplied in Definition 3.1; identifying it requires a further typed map. $\square$

Three zeros must therefore remain distinct:

$$\begin{aligned} 0_F : \quad & O|\psi\rangle = 0 && \text{(formal/operator-relative),} \\ 0_O : \quad & |C(Ox) - q_{\text{ref}}| \leq \delta && \text{(operational),} \\ 0_\Omega : \quad & \text{absolute absence under every registry} && \text{(ontological assertion).} \end{aligned}$$

An experiment can model $0_F \rightarrow 0_O$. No improvement of δ alone provides $0_O \rightarrow 0_\Omega$.

First handwave surface

The chain $N|0\rangle = 0 \Rightarrow$ no quanta of $N \Rightarrow$ empty space $\Rightarrow$ physical nothingness uses one glyph across at least three sorts. The first implication can be definitional; the last two require maps not contained in the eigenvalue equation.

4. Vacuum states, particles, and polarization

For a selected annihilation family $\{a_k\}$, a vacuum state satisfies

$$a_k|0_a\rangle = 0.$$

Under a Bogoliubov transformation

$$b_j = \sum_k (\alpha_{jk} a_k + \beta_{jk} a_k^\dagger),$$

the same state generally has

$$\langle 0_a | N_j^{(b)} | 0_a \rangle = \sum_k |\beta_{jk}|^2, \quad N_j^{(b)} = b_j^\dagger b_j.$$

Thus “no particles” is indexed to an operator family, state representation, and physical setup. This does not make particles or vacua unreal. It prevents an operator-relative zero from serving as absolute emptiness.

Vacuum polarization is equally technical. In schematic momentum-space notation, a dressed propagator has the form

$$D(q) = \frac{D_0(q)}{1 - \Pi(q^2)},$$

with Π a polarization function after the theory’s stated regularization and renormalization. This describes modified propagation or response within the field theory. Translating it as “nothingness becomes

an organized material” changes the sort of the subject. The mathematics did not perform that promotion.

Proposition 4.1 (Lexical paradox manufacture). *If empty means an operator-relative vacuum in the premise and ontological absence in the conclusion, then “empty space contains energy” is not a contradiction derived from one definition. It is an equivocation between registries.*

The correction is not “the vacuum is truly empty” or “the vacuum is truly a substance.” It is to retain the indexing long enough to state what was calculated.

5. The Casimir effect: what closes mathematically

5.1 Ideal parallel plates

For perfect parallel conductors of area A and separation a , a familiar formal assignment is

$$E_0(a) = \frac{\hbar}{2} \sum_{\lambda} \omega_{\lambda}(a),$$

which diverges before a comparison and prescription are specified. The standard finite ideal result is

$$\frac{\Delta E(a)}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \quad P(a) = -\frac{\partial}{\partial a} \frac{\Delta E}{A} = -\frac{\pi^2 \hbar c}{240 a^4}.$$

The measured object is a force or force gradient under finite geometry, material response, temperature, roughness, patch-potential, and calibration models. No experiment separately measures two divergent mode sums and observes their subtraction.

5.2 Scattering and material-response form

At zero temperature, a broad class of planar Casimir–Lifshitz interactions can be written schematically as

$$\frac{E(a)}{A} = \frac{\hbar}{2\pi} \int_0^\infty d\xi \int \frac{d^2 \mathbf{k}_{\parallel}}{(2\pi)^2} \log \det [I - R_1(i\xi, \mathbf{k}_{\parallel}) R_2(i\xi, \mathbf{k}_{\parallel}) e^{-2\kappa a}],$$

where R_1, R_2 encode the bodies’ reflection response and κ the imaginary-frequency propagation factor. The expression makes the relational carrier explicit: two responses, a separation, propagation, and a comparison determinant.

Lifshitz theory extends the ideal result to dispersive materials and intervening media. Experiments in liquids, with semiconductors, corrugations, anisotropic materials, and tailored surfaces confirm that sign and magnitude depend on the coupled material–geometry–temperature registry. This is precisely why the force cannot be assigned to either “empty space” or “the plates” alone without choosing a representation.

5.3 Representation underdetermination

Jaffe showed that Casimir forces can be formulated as relativistic quantum forces between charges and currents without using zero-point energies as independently real sources; the force vanishes with the elec-

tromagnetic coupling. Schwinger's source-theory program likewise avoided making vacuum fluctuations primitive. Milonni and standard field presentations organize the same domain through vacuum fluctuations and correlations. Lifshitz response theory and scattering formulations place material response and Green functions at the center.

These descriptions do not automatically disagree about the measured force. Where they induce the same readout distribution, the Casimir experiment alone cannot choose among their ontological glosses. The calculational equivalence is not proof that the descriptions mean the same thing; it is proof that the selected readout does not carry their difference.

Casimir audit result

$\text{Audit}(\{\Delta E(a), P(a)\}, \text{an absolute extractable vacuum reservoir exists}) = \text{Underdetermined.}$

The force is derived. The reservoir conclusion is neither derived nor contradicted by that result alone.

6. The camps, compared at their actual propositions

The labels below identify recurrent representational commitments, not fixed memberships. Individual authors may occupy more than one row or change position between problems.

Registry family	Primitive emphasis	What its mathematics can carry	Promotion requiring more work
Vacuum-fluctuation / mode	field operators, ground state, correlations, mode spectrum	QED observables; boundary-conditioned correlations; standard Casimir representations	vacuum as independently extractable substance
Source-interaction	charges, currents, propagators, interactions	Casimir and other QED forces without primitive zero-point energy	absence of every meaningful vacuum ontology
Lifshitz / material response	dielectric functions, reflection data, Green functions, temperature	forces for real materials and media; sign and magnitude changes	one uniquely fundamental causal story
Stochastic electrodynamics	classical random zero-point field with Lorentz-invariant spectrum	model-specific fluctuation and response predictions	empirical equivalence to QED in all regimes; available reservoir
Static Casimir engineering	geometry, materials, force landscapes	actuation, stiction control, repulsion under conditions, energy conversion during strokes	positive net work over an unpriced return cycle

Registry family	Primitive emphasis	What its mathematics can carry	Promotion requiring more work
Dynamical Casimir	time-dependent boundary or circuit parameter, pump, output correlations	pump-driven correlated photon production and squeezing	energy appearing without the modulation source
Extraction proposals	nonlinear elements, cavities, gas or atomic cycles	testable device-specific work and heat claims	ZPE attribution without complete ledger and discriminating control
Cosmological vacuum	stress-energy, gravity, renormalized parameters	model-dependent gravitational consequences and cosmological fits	Casimir measurement as direct determination of absolute cosmological vacuum energy
Disclosure-adjacent engineering	commissioned reports, testimony, alleged advanced propulsion	provenance claims and proposed mathematical scenarios	institutional attention or unresolved observation as technical validation
Categorical dismissal	established theory and failed prior devices	strong priors, controls, and known no-go results within scope	impossibility of every undeclared extension

6.1 Vacuum-mode realism and the source-theory reply

Vacuum-fluctuation language provides a powerful, compact QED representation. The error audited here is not its use; it is conversion of representation into sole ontology by rhetorical inheritance. Conversely, Jaffe's interaction formulation blocks the claim that Casimir data uniquely witness zero-point energies, but it does not prove that every vacuum-based representation lacks meaning. Both camps overreach only when calculational sufficiency becomes ontological exclusion.

6.2 Stochastic electrodynamics and extraction

Stochastic electrodynamics assigns a classical random background a spectrum chosen to reproduce selected quantum-like effects. Boyer's work supplies important examples; Puthoff and later authors extend related vocabulary toward gravitation, inertia, or engineering. The registry audit asks, for each claim, which probability law, cutoff, coupling, equilibrium state, and discriminator are supplied. Reproducing one marginal distribution does not establish equality of theories:

$$p(y \mid M_1, \mathcal{E}_1) = p(y \mid M_2, \mathcal{E}_1) \not\Rightarrow p(y \mid M_1, \mathcal{E}) = p(y \mid M_2, \mathcal{E}) \quad \forall \mathcal{E}.$$

Model’s published extraction assessment is useful because it states equilibrium, detailed-balance, and conservation objections while reporting attempted gas-flow experiments. Its conclusion that proposed methods had not been reliably demonstrated is not a theorem that no successor device can work. It is a status claim plus a demand for a complete mechanism and ledger.

6.3 Dynamical Casimir effects

In superconducting circuits, rapid modulation of an effective electrical length produces correlated microwave photons and two-mode squeezing consistent with dynamical-Casimir predictions. The modulation is an explicit energetic input. Describing the output as photons “from the vacuum” can be legitimate inside that field registry; deleting the pump from the engineering ledger is not.

If $H(\lambda(t))$ is driven, then

$$\frac{d}{dt}\langle H \rangle = \left\langle \frac{\partial H}{\partial t} \right\rangle + \text{exchange terms.}$$

The time-dependent control supplies the channel by which energy reaches the output. The experiment demonstrates boundary-conditioned quantum conversion, not unpriced generation.

6.4 UAP and disclosure adjacency

The DIA’s released speculative papers and historical NASA studies show that vacuum and metric-engineering ideas entered government-sponsored exploratory analysis. House hearings and AARO records show ongoing institutional attention to UAP reporting and transparency. The public often combines these facts into a single narrative. The transition

a document exists $\longrightarrow$ its mechanism was demonstrated

is invalid. So is

a report remains unresolved $\longrightarrow$ the preferred propulsion theory explains it.

The proper audit preserves provenance while withholding mechanism assignment until kinematic data, error models, energy signatures, materials, or reproducible hardware discriminate it.

7. Work, cycles, and the missing return path

Theorem 7.1 (Equilibrium one-surface cycle). *Let a quasistatic device be described on a simply connected parameter domain by a differentiable, single-valued equilibrium free energy $\mathcal{F}(\lambda)$, with generalized force*

$$F_\lambda = -\nabla_\lambda \mathcal{F}.$$

For every closed piecewise-smooth cycle Γ in that domain,

$$W_\Gamma = \oint_\Gamma F_\lambda \cdot d\lambda = 0.$$

Proof. The force one-form is exact: $F_\lambda \cdot d\lambda = -d\mathcal{F}$. The integral of an exact form around a closed curve is zero. $\square$

This theorem has explicit scope. Hysteresis, nonequilibrium reservoirs, feedback, switching among free-energy surfaces, nonconservative drives, and topologically obstructed parameter domains can invalidate its premises. They do not invalidate accounting; they identify additional work, heat, state, or path terms.

For an engineering cycle, require

$$W_{\text{net}} = W_{\text{delivered}} - W_{\text{preparation}} - W_{\text{switching}} - W_{\text{reset}} \\ - W_{\text{control}} - W_{\text{measurement}} - W_{\text{parasitic}},$$

with sign conventions and uncertainty stated before the experiment. A positive attractive stroke

$$W_{1 \rightarrow 2} = -[\mathcal{F}(\lambda_2) - \mathcal{F}(\lambda_1)] > 0$$

does not imply $W_{\Gamma} > 0$.

The return-path test

Any proposal that prices the inward Casimir stroke but not separation, boundary restoration, material reset, pump recovery, or thermalization has not yet specified a cycle. Its failure is Underdetermined, not automatically Contradicted: the missing terms may close negatively, at zero, or positively, but the displayed stroke does not decide them.

8. Identifiability: when interpretations cannot be selected

Theorem 8.1 (Readout-equivalence barrier). *Let models M_1, M_2 induce identical probability measures on the observations available under experiment $\mathcal{E}$:*

$$p(y \mid M_1, \mathcal{E}) = p(y \mid M_2, \mathcal{E}) \quad \text{for almost every } y.$$

Then no decision rule based only on repeated samples from that experiment can obtain evidential discrimination between the models.

Proof. For every sample $y_{1:n}$, the likelihood ratio equals one:

$$\Lambda(y_{1:n}) = \prod_{i=1}^n \frac{p(y_i \mid M_1, \mathcal{E})}{p(y_i \mid M_2, \mathcal{E})} = 1.$$

Therefore the data distribution of every statistic is identical under both models; no test has power exceeding its size against this pair. Bayesian posterior odds remain prior odds. $\square$

The theorem does not declare the models identical. It says the chosen projection does not carry their difference. A discriminating experiment must supply $\mathcal{E}^*$ for which

$$p(y \mid M_1, \mathcal{E}^*) \neq p(y \mid M_2, \mathcal{E}^*).$$

This applies in both directions. A Casimir-force curve cannot uniquely certify a vacuum substance when a source-interaction calculation predicts the same curve. Nor can that agreement prove that no future

gravitational, nonequilibrium, or correlation-sensitive experiment could distinguish the registries.

9. A formal taxonomy of handwaving

Code	Unsupported move	Missing object
H1	one glyph used across distinct zero sorts	type declaration and coercion
H2	formal vacuum changed into ontological emptiness	state-to-ontology transition map
H3	finite difference promoted to absolute energy substance	absolute comparator and coupling model
H4	positive stroke promoted to positive cycle	return path and full ledger
H5	a fitting interpretation promoted to uniquely confirmed mechanism	discriminating experiment
H6	successful model closure promoted to ontological exhaustion	proof excluding all relevant extensions
H7	unresolved observation promoted to preferred technology	likelihood-bearing data and comparator
H8	institutional attention promoted to validation	independent technical evidence
H9	limit notation promoted to completed witness of zero or infinity	attainability and registry-independent identification
H10	anomaly assigned to the desired reservoir	calibrated residual model and alternative-account tests

These codes are not verdicts about character or intelligence. They name repairable proof obligations.

Definition 9.1 (Inferential gap). For premises Γ and conclusion C , define

$$\mathcal{G}(\Gamma, C) = \{\Delta : \Gamma \cup \Delta \vdash C\}.$$

A minimal gap completion Δ^* satisfies $\Gamma \cup \Delta^* \vdash C$, while no proper subset does.

The method speaks where enthusiasm or authority attempts to pressure across a gap because it turns the gap into a work order. For a ZPE device, Δ^* may contain a controllable state variable, coupling law, nonequilibrium preparation, closed cycle, complete ledger, uncertainty model, and distinguishing readout. Naming those objects advances the proposal. Repeating the verdict does not.

10. Proof-carrying specification for ZPE engineering

Every device claim should publish the tuple

$$\mathcal{D} = (\mathcal{R}, B, \lambda, \Pi, \Gamma, \mathcal{L}, \mathcal{E}, \mathcal{K}, \mathcal{S}),$$

where $\mathcal{R}$ is the registry; B the boundary configuration; λ the controls; Π the preparation protocol; Γ the full cycle; $\mathcal{L}$ the energy/entropy ledger; $\mathcal{E}$ the measurement model; $\mathcal{K}$ the control comparators; and $\mathcal{S}$ the stopping rule.

10.1 Minimum experimental record

1. Declare every use of zero: electrical offset, baseline subtraction, ground-state index, null count, or thermodynamic reference.
2. Draw the entire state cycle, including preparation and reset.
3. Measure control-channel work at the device boundary, not only nominal source settings.
4. Report heat, stored-energy changes, hysteresis, drift, material relaxation, and switching transients.
5. Separate preregistered primary readouts from exploratory diagnostics.
6. Use independently calibrated controls for electrostatic patches, vibration, RF pickup, thermal gradients, chemistry, and data-selection effects.
7. Specify a prediction on which the ZPE account and strongest ordinary comparator differ.
8. Blind analysis where discretionary selection can manufacture residue.
9. Propagate uncertainty through the net-cycle quantity.
10. State which result would revise the mechanism claim.

An evidential residue can be written

$$R = W_{\text{delivered}} - \sum_i W_i - \sum_j Q_j - \Delta U,$$

with covariance-aware uncertainty σ_R . A nonzero R/σ_R is not yet ZPE; it is a failure of the present ledger or model. Attribution requires the ZPE model to predict the residual's dependence on controls where alternatives do not.

No privileged anomaly

The registry method protects speculation by refusing premature dismissal. It also protects the experiment by refusing to let an unexplained residual inherit the preferred ontology. Residue is a location for the next comparison, not a substance discovered by subtraction.

11. Relation to nested transfer and the Infinite Registry

The companion paper *Nested Transfer and Scale-Free Spectra* studies readouts formed by a continuum, an effective weighting, finite boundaries, and a projection. Casimir–Lifshitz expressions admit a precise structural correspondence:

$$S(f) = \int K(f, \tau) g(\tau) d\tau,$$

and

$$\frac{E(a)}{A} = \int_0^\infty d\xi \int d^2 k_{\parallel} \mathcal{K}(\xi, k_{\parallel}; a, R_1, R_2)$$

both register a continuum through a weighted, boundary-conditioned readout. This is a Tier-II correspondence. It does not make trap noise and Casimir physics one mechanism.

The Infinite Registry adds the stopping discipline. Zero and infinity are not completed objects awaiting discovery at opposite ends of a line. Zero is registered through equivalence, calibration, and gate; infinity through an indefinitely extensible rule. A finite calculation can close under its assumptions while leaving its frame edge explicit. The ability to extend the registry does not prove that no outside exists; inability to name an outside from the current language does not turn that outside into a witnessed higher object.

Applied to ZPE, this means that “aether” or “fundamental medium” may be introduced as a primitive, but the introduction does not close explanation. It opens a successor registry requiring states, transformations, coupling laws, conserved quantities, and discriminating readouts:

$$\mathcal{R}' = \mathcal{R} \cup \{\text{medium primitive and its rules}\}.$$

The medium may prove useful. The label alone carries no residue across the transition.

12. What is claimed, and what is not

Claimed at Tier I.

1. Cross-sort substitution requires a declared type conversion and predicate preservation (Theorem 2.2).
2. Refinement of null acceptance sets alone does not establish an attainable, registry-independent absolute zero (Proposition 3.2).
3. A conservative equilibrium force on one free-energy surface yields zero work around a closed cycle (Theorem 7.1).
4. Readout-equivalent models cannot be discriminated by repetition of that readout (Theorem 8.1).
5. Standard Casimir and dynamical-Casimir observations establish the scoped observables described by their experimental models; they do not logically entail every ontology attached to their vocabulary.

Claimed at Tier II. The recurring ZPE disputes can be read as registry transitions in which lexical identity hides changes of mathematical object. The audit taxonomy locates these transitions without installing one representation as fundamental.

Not claimed.

- That vacuum-state, source-interaction, Lifshitz, or stochastic representations are meaningless.
- That every proposed ZPE device must fail, or that any has demonstrated net extraction.
- That the Casimir effect is unreal, merely semantic, or exhausted by ideal plates.
- That official interest verifies extraordinary technology, or that unresolved UAP reports are unimportant.
- That conservation laws may be ignored where the current ledger is incomplete.
- That the registry calculus supplies a completed ontology of reality.

13. Falsification and defeat surfaces

F1 — formal surface. Exhibit an invalid step in Theorems 2.2, 7.1, or 8.1, or in Proposition 3.2 under their stated assumptions.

F2 — classification surface. Exhibit a named application classified as ill-typed or underdetermined for which the cited proponent actually supplies the required map, cycle, or discriminator.

F3 — cycle surface. Demonstrate a reproducible closed device cycle with positive net work after all declared inputs, heat flows, stored-energy changes, preparation, switching, control, and reset terms are independently measured within uncertainty.

F4 — attribution surface. For an observed residual, preregister a dependence uniquely predicted by the ZPE mechanism among the compared models and reproduce it under blinded controls.

F5 — zero surface. Supply an operational procedure whose output establishes registry-independent ontological absence, rather than another calibrated null acceptance set, and state the proof of that transport.

F6 — institutional surface. Supply an official source that directly verifies the extraordinary mechanism attributed to it, rather than documenting study, allegation, unresolved observation, or speculative analysis.

F3 and F4 would not defeat the formal calculus; they would satisfy its work order and defeat a negative status assessment. This asymmetry is intentional. The method must permit evidence to change the ledger.

14. Scoped conclusion

Zero is indispensable because comparison requires an origin. Its usefulness also conceals its construction. The same symbol can name a number-operator eigenvalue, detector acceptance set, renormalization convention, thermodynamic reference, or metaphysical absence. When those meanings are collated, a paradox appears and is then offered as explanation: empty space contains energy; nothing fluctuates; the vacuum becomes polarized; a measured force witnesses an unlimited reservoir.

The mathematics carries narrower and more valuable statements. Quantum theories define ground states and correlations. Casimir–Lifshitz theory predicts boundary- and material-conditioned forces. Dynamical modulation converts supplied work into correlated radiation under a quantum description. None of these results becomes smaller when stripped of ontological overloading. They become testable at the exact surface where their evidence lies.

The registry method therefore does not settle the vacuum by decree. It prevents either enthusiasm or authority from crossing an inferential gap without carrying the missing object. Where the map exists, use it. Where the cycle closes, measure it. Where models differ, build the discriminator. Where they do not, admit that the selected readout cannot adjudicate their difference.

Closing statement

A description can close a calculation without closing reality. Its incompleteness does not license insertion of a preferred substance; its success does not license exclusion of every successor registry. Zero is a coordinate of a declared relation. It is not nothing merely because the glyph is empty, and it is not a reservoir merely because the ground state is nontrivial.

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